

investors to exchange these assets against the current bid–ask on the order book and uses a hybrid on–off chain approach. Essentially, dYdX stores *signed*, or preapproved, orders without submitting to Ethereum; they use cryptography to guarantee they are used only to exchange funds for the desired asset at the desired price. The DEX supports limit orders and a *maximum slippage* parameter for market orders in an effort to mitigate the slippage associated with price moves or front-running.

dYdX provides market makers and traders the open-source software and a user interface required to interact with the DEX. Having dYdX do the order matching introduces a certain element of trust because the infrastructure could be in downtime or not posting transactions for some reason. Allowing dYdX to match the orders holds little or no risk that the company could steal user funds because the signed orders can be used only as intended per the smart contract. When the orders are matched, they are submitted to the Ethereum blockchain, where the smart contract facilitates settlement.

In addition, an investor can take a levered long or short position up to 10 times using margined collateral. The positions can be isolated so that a single collateral deposit is used or cross-margined to pool investor's balances for collateral. As in other protocols, dYdX has a maintenance margin requirement that, if not maintained, triggers liquidation of the collateral to close the position. The liquidations can be performed by external keepers who are paid to find and